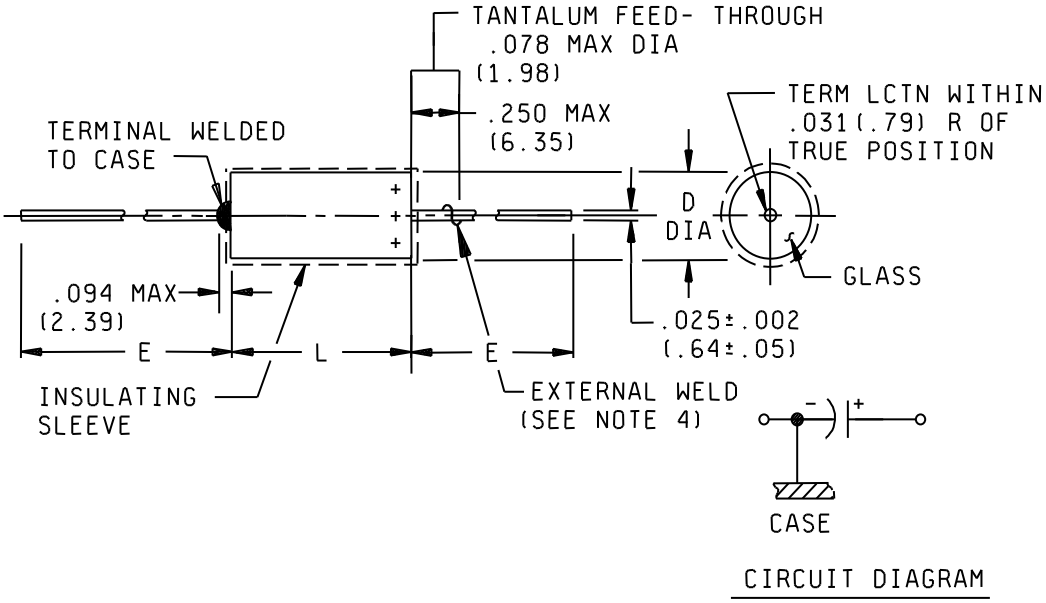
INCH-POUND
MIL-PRF-39006/31D
21 March 2018
SUPERSEDING
MIL-PRF-39006/31C
21 April 2011

PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM (POLARIZED, SINTERED SLUG), EXTENDED RANGE, +85°C (VOLTAGE DERATED TO +125°C), ESTABLISHED RELIABILITY, STYLE CLR91

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39006.




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR91 capacitors.



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| Case size | Dimensions in inches (mm) | | | |
|---|---|---|---|---|
| | Basic case | | Insulated case | |
| | L 1/ +.031 (0.79) -.016 (0.41) | D ±.016 (0.41) | D max | E ±.250 (6.35) |
| T1 | .453 (11.51) | .188 (4.78) | .219 ( 5.56) | 1.500 (38.10) |
| T2 | .641 (16.28) | .281 (7.14) | .312 ( 7.92) | 2.250 (57.15) |
| T3 | .766 (19.46) | .375 (9.52) | .406 (10.31) | 2.250 (57.15) |
| T4 | 1.062 (26.97) | .375 (9.52) | .406 (10.31) | 2.250 (57.15) |

1/ Length of basic case sleeving shall be as specified in MIL-PRF-39006.

FIGURE 1. Style CLR91 capacitors - Continued.

QUALIFICATION:

A manufacturer shall be considered qualified to this specification sheet by similarity if the manufacturer is qualified to style CLR81 capacitors in MIL-PRF-39006/25 and if the design and construction of the capacitors in this specification sheet is the same as the design and construction of the capacitors covered by MIL-PRF-39006/25.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Tantalum, tubular, insulated.

Seal type: Hermetic (glass-to-tantalum anode).

Terminals: Axial-wire lead: Cathode and anode (type N32, N51, or N52 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

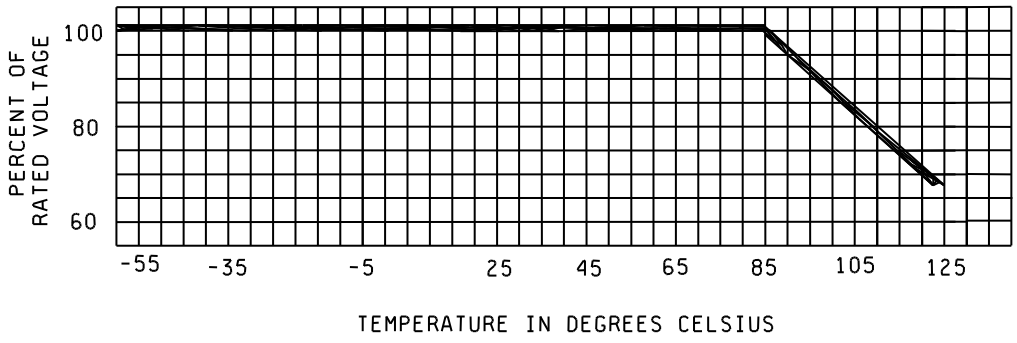



FIGURE 2. Voltage derating with temperature.



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Rated voltage: See table I.

Capacitance (Cap.) nominal (nom): See table I.

Cap. tolerance: ±10 percent or ±20 percent (see table I).

Failure rate level: M, P, R, and S (see table I).

DC leakage (DCL) (max) (at +25°C): See table I.

Cap. (nominal): MIL-STD-202-305.

Dissipation factor (DF) (max): See table I.

Seal: MIL-STD-202-112, conditions A or D, and C.

Shock (specified pulse): MIL-STD-202-213, condition I (100 g's) or condition D (500 g's), as applicable.

Vibration, high frequency: MIL-STD-202-204, condition D (20 g's) or condition H (80 g's), as applicable.

Random vibration (when applicable): MIL-STD-202-214, test condition II-K (53.79 g's).

Thermal shock: MIL-STD-202-107, condition A (with step 3 at +125°C). The following shall apply:

Number of cycles: In accordance with MIL-PRF-39006.

DCL: 200 percent (max) of +25°C value of table I for qualification and group C.

DCL: 125 percent (max) of +25°C value of table I for group B.

ΔCap: Within ±5 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Salt spray (corrosion): MIL-STD-202-101, condition B (48 hours).

Solderability: MIL-STD-202-208.

Terminal strength:

Pull: MIL-STD-202-211, condition A.

Wire-lead bend: In accordance with MIL-PRF-39006.

Surge voltage: See table I.

Moisture resistance: MIL-STD-202-106.

DCL: 125 percent (max) of +25°C value of table I.

ΔCap: Within ±8 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Dielectric withstanding voltage (DWV): MIL-STD-202-301. 2,000 V dc, minimum.

Insulation resistance (IR): MIL-STD-202-302, condition B. 100 megohms, minimum.



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Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Impedance. See table I.

ΔCap: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.



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Reverse voltage: As specified in MIL-PRF-39006.

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Life at +85°C: MIL-STD-202-108.

2,000 hours (qualification):

DCL: See table I.

ΔCap: Within ± 10 percent of initial measurement.

DF: See table I.

10,000 hours (group B) and extended life:

DCL (+85°C): 125 percent (max) of table I value.

DCL (+25°C): See table I.

ΔCap: Within +10 percent, -20 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

AC ripple life at +85°C (qualification and group C):

Test ripple current: See table I.

Test voltage: 66.67 percent of rated voltage (see table I).

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

Life at +125°C: MIL-STD-202-108.

DCL: See table I.

ΔCap: Within ± 10 percent of initial measurement.

DF: See table I.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:20Z
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2,000 hours (group C):

DCL (+125°C): 125 percent (max) of table I value.

DCL (+25°C): See table I.

ΔCap: Within +10 percent, -20 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

Barometric pressure: MIL-STD-202-105, condition E (150,000 feet) (45,720.1 m).

Test voltage: Rated voltage.

Resistance to soldering heat: MIL-STD-202-210, condition C.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Marking: In accordance with MIL-PRF-39006, with the following exception: The manufacturer may mark the T1 case size "H" designated capacitors as follows:

Polarity of all parts: Series of (+) symbols encircling positive end (see figure 1).

Example: M39006/
31H0001
JAN TM
12345
9415A

Voltage groups (+85°C rated voltage):

I - - - - - - - - - - 6 V through 15 V

II - - - - - - - - - - 25 V through 75 V.

III - - - - - - - - - - 100 V and 125 V.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:20Z
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TABLE I. Capacitor characteristics.

| Cap. | Cap. tolerance | DC leakage (max) | | Dissipation factor (max) | ESR at +25°C (max) 2/ | Impedance (max) | Capacitance change at | | | Max +85°C 40 kHz ripple current 3/ | Case size | PIN M39006/31- 4/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | +25°C | +85°C +125°C | | | | -55°C | +85°C | +125°C | | | Failure rate level (%/1,000 hrs) | | | |
| | | | | | | | | | | | | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| µF | ±% | µA | µA | % | Ohms | Ohms | % | % | % | mA rms | | | | | |
| 6 $V_{dc}$ at +85°C 1/ | | | | | | | 4 $V_{dc}$ at +125°C 1/ | | | | Surge (+85°C) 6.9 $V_{dc}$ | | | | |
| 220 | ±20 | 2 | 9 | 25 | 1.51 | 36 | -64 | +13 | +16 | 1,000 | T1 | 0001- | 0089- | 0177- | 0265- |
| 220 | ±10 | 2 | 9 | 25 | 1.51 | 36 | -64 | +13 | +16 | 1,000 | T1 | 0002- | 0090- | 0178- | 0266- |
| 820 | ±20 | 3 | 14 | 77.5 | 1.26 | 18 | -88 | +16 | +20 | 1,500 | T2 | 0003- | 0091- | 0179- | 0267- |
| 820 | ±10 | 3 | 14 | 77.5 | 1.26 | 18 | -88 | +16 | +20 | 1,500 | T2 | 0004- | 0092- | 0180- | 0268- |
| 1,500 | ±20 | 5 | 20 | 86 | 0.76 | 18 | -90 | +20 | +25 | 1,900 | T3 | 0005- | 0093- | 0181- | 0269- |
| 1,500 | ±10 | 5 | 20 | 86 | 0.76 | 18 | -90 | +20 | +25 | 1,900 | T3 | 0006- | 0094- | 0182- | 0270- |
| 2,200 | ±20 | 6 | 24 | 85 | 0.52 | 13 | -90 | +25 | +30 | 2,300 | T4 | 0007- | 0095- | 0183- | 0271- |
| 2,200 | ±10 | 6 | 24 | 85 | 0.52 | 13 | -90 | +25 | +30 | 2,300 | T4 | 0008- | 0096- | 0184- | 0272- |
| 8 $V_{dc}$ at +85°C 1/ | | | | | | | 5 $V_{dc}$ at +125°C 1/ | | | | Surge (+85°C) 9.2 $V_{dc}$ | | | | |
| 180 | ±20 | 2 | 9 | 20.5 | 1.51 | 45 | -60 | +13 | +16 | 1,000 | T1 | 0009- | 0097- | 0185- | 0273- |
| 180 | ±10 | 2 | 9 | 20.5 | 1.51 | 45 | -60 | +13 | +16 | 1,000 | T1 | 0010- | 0098- | 0186- | 0274- |
| 680 | ±20 | 3 | 14 | 65 | 1.27 | 22 | -83 | +16 | +20 | 1,500 | T2 | 0011- | 0099- | 0187- | 0275- |
| 680 | ±10 | 3 | 14 | 65 | 1.27 | 22 | -83 | +16 | +20 | 1,500 | T2 | 0012- | 0100- | 0188- | 0276- |
| 1,500 | ±20 | 5 | 20 | 85 | 0.75 | 18 | -90 | +20 | +25 | 1,900 | T3 | 0013- | 0101- | 0189- | 0277- |
| 1,500 | ±10 | 5 | 20 | 85 | 0.75 | 18 | -90 | +20 | +25 | 1,900 | T3 | 0014- | 0102- | 0190- | 0278- |
| 1,800 | ±20 | 7 | 25 | 69 | 0.51 | 14 | -90 | +25 | +30 | 2,300 | T4 | 0015- | 0103- | 0191- | 0279- |
| 1,800 | ±10 | 7 | 25 | 69 | 0.51 | 14 | -90 | +25 | +30 | 2,300 | T4 | 0016- | 0104- | 0192- | 0280- |
| 10 $V_{dc}$ at +85°C 1/ | | | | | | | 7 $V_{dc}$ at +125°C 1/ | | | | Surge (+85°C) 11.5 $V_{dc}$ | | | | |
| 150 | ±20 | 2 | 9 | 17 | 1.51 | 54 | -55 | +13 | +16 | 900 | T1 | 0017- | 0105- | 0193- | 0281- |
| 150 | ±10 | 2 | 9 | 17 | 1.51 | 54 | -55 | +13 | +16 | 900 | T1 | 0018- | 0106- | 0194- | 0282- |
| 560 | ±20 | 3 | 16 | 53 | 1.26 | 27 | -77 | +16 | +20 | 1,450 | T2 | 0019- | 0107- | 0195- | 0283- |
| 560 | ±10 | 3 | 16 | 53 | 1.26 | 27 | -77 | +16 | +20 | 1,450 | T2 | 0020- | 0108- | 0196- | 0284- |
| 1,200 | ±20 | 5 | 20 | 68.5 | 0.76 | 18 | -88 | +20 | +25 | 1,850 | T3 | 0021- | 0109- | 0197- | 0285- |
| 1,200 | ±10 | 5 | 20 | 68.5 | 0.76 | 18 | -88 | +20 | +25 | 1,850 | T3 | 0022- | 0110- | 0198- | 0286- |
| 1,500 | ±20 | 7 | 25 | 57 | 0.51 | 15 | -88 | +25 | +30 | 2,300 | T4 | 0023- | 0111- | 0199- | 0287- |
| 1,500 | ±10 | 7 | 25 | 57 | 0.51 | 15 | -88 | +25 | +30 | 2,300 | T4 | 0024- | 0112- | 0200- | 0288- |
| 15 $V_{dc}$ at +85°C 1/ | | | | | | | 10 $V_{dc}$ at +125°C 1/ | | | | Surge (+85°C) 17.2 $V_{dc}$ | | | | |
| 100 | ±20 | 2 | 9 | 15 | 1.99 | 72 | -44 | +13 | +16 | 900 | T1 | 0025- | 0113- | 0201- | 0289- |
| 100 | ±10 | 2 | 9 | 15 | 1.99 | 72 | -44 | +13 | +16 | 900 | T1 | 0026- | 0114- | 0202- | 0290- |
| 390 | ±20 | 3 | 16 | 37 | 1.26 | 31 | -66 | +16 | +20 | 1,450 | T2 | 0027- | 0115- | 0203- | 0291- |
| 390 | ±10 | 3 | 16 | 37 | 1.26 | 31 | -66 | +16 | +20 | 1,450 | T2 | 0028- | 0116- | 0204- | 0292- |
| 820 | ±20 | 6 | 24 | 55.5 | 0.90 | 22 | -77 | +20 | +25 | 1,800 | T3 | 0029- | 0117- | 0205- | 0293- |
| 820 | ±10 | 6 | 24 | 55.5 | 0.90 | 22 | -77 | +20 | +25 | 1,800 | T3 | 0030- | 0118- | 0206- | 0294- |
| 1,000 | ±20 | 8 | 32 | 46 | 0.61 | 17 | -77 | +25 | +30 | 2,300 | T4 | 0031- | 0119- | 0207- | 0295- |
| 1,000 | ±10 | 8 | 32 | 46 | 0.61 | 17 | -77 | +25 | +30 | 2,300 | T4 | 0032- | 0120- | 0208- | 0296- |
| 25 $V_{dc}$ at +85°C 1/ | | | | | | | 15 $V_{dc}$ at +125°C 1/ | | | | Surge (+85°C) 28.8 $V_{dc}$ | | | | |
| 68 | ±20 | 2 | 9 | 11 | 2.15 | 90 | -40 | +12 | +15 | 850 | T1 | 0033- | 0121- | 0209- | 0297- |
| 68 | ±10 | 2 | 9 | 11 | 2.15 | 90 | -40 | +12 | +15 | 850 | T1 | 0034- | 0122- | 0210- | 0298- |
| 270 | ±20 | 3 | 16 | 27.5 | 1.35 | 33 | -62 | +13 | +16 | 1,400 | T2 | 0035- | 0123- | 0211- | 0299- |
| 270 | ±10 | 3 | 16 | 27.5 | 1.35 | 33 | -62 | +13 | +16 | 1,400 | T2 | 0036- | 0124- | 0212- | 0300- |
| 560 | ±20 | 7 | 28 | 38 | 0.90 | 24 | -72 | +20 | +25 | 1,750 | T3 | 0037- | 0125- | 0213- | 0301- |
| 560 | ±10 | 7 | 28 | 38 | 0.90 | 24 | -72 | +20 | +25 | 1,750 | T3 | 0038- | 0126- | 0214- | 0302- |
| 680 | ±20 | 8 | 32 | 31.5 | 0.62 | 19 | -72 | +25 | +30 | 2,100 | T4 | 0039- | 0127- | 0215- | 0303- |
| 680 | ±10 | 8 | 32 | 31.5 | 0.62 | 19 | -72 | +25 | +30 | 2,100 | T4 | 0040- | 0128- | 0216- | 0304- |
| 30 $V_{dc}$ at +85°C 1/ | | | | | | | 20 $V_{dc}$ at +125°C 1/ | | | | Surge (+85°C) 34.5 $V_{dc}$ | | | | |
| 56 | ±20 | 2 | 9 | 11 | 2.61 | 100 | -38 | +12 | +15 | 800 | T1 | 0041- | 0129- | 0217- | 0305- |
| 56 | ±10 | 2 | 9 | 11 | 2.61 | 100 | -38 | +12 | +15 | 800 | T1 | 0042- | 0130- | 0218- | 0306- |
| 220 | ±20 | 3 | 16 | 21 | 1.27 | 36 | -60 | +13 | +16 | 1,200 | T2 | 0043- | 0131- | 0219- | 0307- |

See footnotes at the end of table.



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TABLE I. Capacitor characteristics - Continued.

| Cap. | Cap. tolerance | DC leakage (max) | | Dissipation factor (max) | ESR at +25°C (max) 2/ | Impedance (max) | Capacitance change at | | | Max +85°C 40 kHz ripple current 3/ | Case size | PIN M39006/31- 4/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | +25°C | +85°C +125°C | | | | -55°C | +85°C | +125°C | | | Failure rate level (%/1,000 hrs) | | | |
| µF | ±% | µA | µA | % | Ohms | Ohms | % | % | % | mA rms | | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| 30 $V_{dc}$ at +85°C 1/ | | | | | | | 20 $V_{dc}$ at +125°C 1/ | | | | | Surge (+85°C) 34.5 $V_{dc}$ -Continued. | | | |
| 220 | ±10 | 3 | 16 | 21 | 1.27 | 36 | -60 | +13 | +16 | 1,200 | T2 | 0044- | 0132- | 0220- | 0308- |
| 470 | ±20 | 8 | 32 | 32 | 0.91 | 25 | -65 | +20 | +25 | 1,500 | T3 | 0045- | 0133- | 0221- | 0309- |
| 470 | ±10 | 8 | 32 | 32 | 0.91 | 25 | -65 | +20 | +25 | 1,500 | T3 | 0046- | 0134- | 0222- | 0310- |
| 560 | ±20 | 9 | 36 | 27.5 | 0.65 | 20 | -65 | +25 | +30 | 2,000 | T4 | 0047- | 0135- | 0223- | 0311- |
| 560 | ±10 | 9 | 36 | 27.5 | 0.65 | 20 | -65 | +25 | +30 | 2.000 | T4 | 0048- | 0136- | 0224- | 0312- |
| 50 $V_{dc}$ at +85°C 1/ | | | | | | | 30 $V_{dc}$ at +125°C 1/ | | | | | Surge (+85°C) 57.5 $V_{dc}$ | | | |
| 33 | ±20 | 2 | 9 | 6.15 | 2.48 | 135 | -29 | +10 | +12 | 700 | T1 | 0049- | 0137- | 0225- | 0313- |
| 33 | ±10 | 2 | 9 | 6.15 | 2.48 | 135 | -29 | +10 | +12 | 700 | T1 | 0050- | 0138- | 0226- | 0314- |
| 120 | ±20 | 4 | 24 | 11.25 | 1.25 | 49 | -42 | +12 | +15 | 1,200 | T2 | 0051- | 0139- | 0227- | 0315- |
| 120 | ±10 | 4 | 24 | 11.25 | 1.25 | 49 | -42 | +12 | +15 | 1,200 | T2 | 0052- | 0140- | 0228- | 0316- |
| 270 | ±20 | 8 | 32 | 18.5 | 0.91 | 29 | -46 | +20 | +25 | 1,450 | T3 | 0053- | 0141- | 0229- | 0317- |
| 270 | ±10 | 8 | 32 | 18.5 | 0.91 | 29 | -46 | +20 | +25 | 1,450 | T3 | 0054- | 0142- | 0230- | 0318- |
| 330 | ±20 | 9 | 36 | 19 | 0.77 | 22 | -46 | +25 | +30 | 1,900 | T4 | 0055- | 0143- | 0231- | 0319- |
| 330 | ±10 | 9 | 36 | 19 | 0.77 | 22 | -46 | +25 | +30 | 1,900 | T4 | 0056- | 0144- | 0232- | 0320- |
| 60 $V_{dc}$ at +85°C 1/ | | | | | | | 40 $V_{dc}$ at +125°C 1/ | | | | | Surge (+85°C) 69.0 $V_{dc}$ | | | |
| 27 | ±20 | 3 | 12 | 5.1 | 2.51 | 144 | -24 | +10 | +12 | 700 | T1 | 0057- | 0145- | 0233- | 0321- |
| 27 | ±10 | 3 | 12 | 5.1 | 2.51 | 144 | -24 | +10 | +12 | 700 | T1 | 0058- | 0146- | 0234- | 0322- |
| 100 | ±20 | 4 | 20 | 9.5 | 1.26 | 54 | -36 | +12 | +15 | 1.100 | T2 | 0059- | 0147- | 0235- | 0323- |
| 100 | ±10 | 4 | 20 | 9.5 | 1.26 | 54 | -36 | +12 | +15 | 1,100 | T2 | 0060- | 0148- | 0236- | 0324- |
| 220 | ±20 | 8 | 32 | 15 | 0.91 | 29 | -40 | +16 | +20 | 1,400 | T3 | 0061- | 0149- | 0237- | 0325- |
| 220 | ±10 | 8 | 32 | 15 | 0.91 | 29 | -40 | +16 | +20 | 1,400 | T3 | 0062- | 0150- | 0238- | 0326- |
| 270 | ±20 | 9 | 36 | 13.5 | 0.67 | 23 | -45 | +20 | +25 | 1,850 | T4 | 0063- | 0151- | 0239- | 0327- |
| 270 | ±10 | 9 | 36 | 13.5 | 0.67 | 23 | -45 | +20 | +25 | 1,850 | T4 | 0064- | 0152- | 0240- | 0328- |
| 75 $V_{dc}$ at +85°C 1/ | | | | | | | 50 $V_{dc}$ at +125°C 1/ | | | | | Surge (+85°C) 86.2 $V_{dc}$ | | | |
| 22 | ±20 | 3 | 12 | 4.25 | 2.57 | 157 | -19 | +10 | +12 | 600 | T1 | 0065- | 0153- | 0241- | 0329- |
| 22 | ±10 | 3 | 12 | 4.25 | 2.57 | 157 | -19 | +10 | +12 | 600 | T1 | 0066- | 0154- | 0242- | 0330- |
| 82 | ±20 | 4 | 24 | 7.6 | 1.23 | 63 | -30 | +12 | +15 | 1,000 | T2 | 0067- | 0155- | 0243- | 0331- |
| 82 | ±10 | 4 | 24 | 7.6 | 1.23 | 63 | -30 | +12 | +15 | 1,000 | T2 | 0068- | 0156- | 0244- | 0332- |
| 180 | ±20 | 9 | 36 | 12.2 | 0.90 | 30 | -35 | +16 | +20 | 1,300 | T3 | 0069- | 0157- | 0245- | 0333- |
| 180 | ±10 | 9 | 36 | 12.2 | 0.90 | 30 | -35 | +16 | +20 | 1,300 | T3 | 0070- | 0158- | 0246- | 0334- |
| 220 | ±20 | 10 | 40 | 18.5 | 1.12 | 24 | -40 | +20 | +25 | 1,800 | T4 | 0071- | 0159- | 0247- | 0335- |
| 220 | ±10 | 10 | 40 | 18.5 | 1.12 | 24 | -40 | +20 | +25 | 1,800 | T4 | 0072- | 0160- | 0248- | 0336- |
| 100 $V_{dc}$ at +85°C 1/ | | | | | | | 65 $V_{dc}$ at +125°C 1/ | | | | | Surge (+85°C) 115.0 $V_{dc}$ | | | |
| 10 | ±20 | 3 | 12 | 2.25 | 2.99 | 200 | -17 | +10 | +12 | 800 | T1 | 0073- | 0161- | 0249- | 0337- |
| 10 | ±10 | 3 | 12 | 2.25 | 2.99 | 200 | -17 | +10 | +12 | 800 | T1 | 0074- | 0162- | 0250- | 0338- |
| 39 | ±20 | 5 | 24 | 5.2 | 1.77 | 80 | -20 | +12 | +15 | 1,300 | T2 | 0075- | 0163- | 0251- | 0339- |
| 39 | ±10 | 5 | 24 | 5.2 | 1.77 | 80 | -20 | +12 | +15 | 1,300 | T2 | 0076- | 0164- | 0252- | 0340- |
| 68 | ±20 | 10 | 40 | 5.65 | 1.11 | 40 | -30 | +14 | +16 | 1,600 | T3 | 0077- | 0165- | 0253- | 0341- |
| 68 | ±10 | 10 | 40 | 5.65 | 1.11 | 40 | -30 | +14 | +16 | 1,600 | T3 | 0078- | 0166- | 0254- | 0342- |
| 120 | ±20 | 12 | 48 | 12.5 | 1.38 | 30 | -35 | +15 | +17 | 2,000 | T4 | 0079- | 0167- | 0255- | 0343- |
| 120 | ±10 | 12 | 48 | 12.5 | 1.38 | 30 | -35 | +15 | +17 | 2,000 | T4 | 0080- | 0168- | 0256- | 0344- |
| 125 $V_{dc}$ at +85°C 1/ | | | | | | | 85 $V_{dc}$ at +125°C 1/ | | | | | Surge (+85°C) 144.0 $V_{dc}$ | | | |
| 6.8 | ±20 | 3 | 12 | 3 | 5.86 | 300 | -14 | +10 | +12 | 700 | T1 | 0081- | 0169- | 0257- | 0345- |
| 6.8 | ±10 | 3 | 12 | 3 | 5.86 | 300 | -14 | +10 | +12 | 700 | T1 | 0082- | 0170- | 0258- | 0346- |
| 27 | ±20 | 5 | 24 | 3.6 | 1.77 | 90 | -18 | +12 | +15 | 1,200 | T2 | 0083- | 0171- | 0259- | 0347- |
| 27 | ±10 | 5 | 24 | 3.6 | 1.77 | 90 | -18 | +12 | +15 | 1,200 | T2 | 0084- | 0172- | 0260- | 0348- |
| 47 | ±20 | 10 | 40 | 3.95 | 1.12 | 50 | -26 | +14 | +16 | 1,500 | T3 | 0085- | 0173- | 0261- | 0349- |

See footnotes at the end of table.



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TABLE I. Capacitor characteristics - Continued.

| Cap. | Cap. tolerance | DC leakage (max) | | Dissipation factor (max) | ESR at +25°C (max) 2/ | Impedance (max) | Capacitance change at | | | Max +85°C 40 kHz ripple current 3/ | Case size | PIN M39006/31- 4/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | +25°C | +85°C +125°C | | | | -55°C | +85°C | +125°C | | | Failure rate level (%/1,000 hrs) | | | |
| | | | | | | | | | | | | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| µF | ±% | µA | µA | % | Ohms | Ohms | % | % | % | mA rms | | | | | |
| 125 $V_{dc}$ at +85°C 1/ | | | | | | | 85 $V_{dc}$ at +125°C 1/ | | | | Surge (+85°C) 144.0 $V_{dc}$ -Continued. | | | | |
| 47 | ±10 | 10 | 40 | 3.95 | 1.12 | 50 | -26 | +14 | +16 | 1,500 | T3 | 0086- | 0174- | 0262- | 0350- |
| 82 | ±20 | 12 | 48 | 8.7 | 1.41 | 32 | -30 | +15 | +17 | 1,900 | T4 | 0087- | 0175- | 0263- | 0351- |
| 82 | ±10 | 12 | 48 | 8.7 | 1.41 | 32 | -30 | +15 | +17 | 1,900 | T4 | 0088- | 0176- | 0264- | 0352- |

1/ Reverse voltage rating at +85°C is 3 V dc and at +125°C is 2 V dc.
2/ Maximum ESR is calculated by the following equation:

$$\text{ESR (max)} = \frac{DF}{2\pi f C}, \text{ where:}$$

DF = Maximum dissipation factor from table I;
f = 120 Hz;
C = Nominal capacitance.

3/ For ripple current limits at various temperatures, voltages, and frequencies, see table II.
4/ Dash number shall include the letter "H" to indicate the optional vibration and shock requirements (i.e., 53.79 g's random vibration, 80 g's sinusoidal vibration, and 500 g's shock) or the "-" will be deleted. The acquiring activity may substitute an "H" designated part in place of a non "H" designated part at their option.

TABLE II. Ripple current multipliers versus frequency, temperature, and applied peak voltage. 1/ 2/ 3/ 4/ 5/

| Frequency of applied ripple current | | 120 Hz | | | | 800 Hz | | | | 1 kHz | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ambient still air temperature in °C | | ≤+55 | +85 | +105 | +125 | ≤+55 | +85 | +105 | +125 | ≤+55 | +85 | +105 | +125 |
| % of | 100% | .60 | .39 | - | - | .71 | .43 | - | - | .72 | .45 | - | - |
| +85°C | 90% | .60 | .46 | - | - | .71 | .55 | - | - | .72 | .55 | - | - |
| rated | 80% | .60 | .52 | .35 | - | .71 | .62 | .42 | - | .72 | .62 | .42 | - |
| peak | 70% | .60 | .58 | .44 | - | .71 | .69 | .52 | - | .72 | .70 | .52 | - |
| voltage | 66-2/3% | .60 | .60 | .46 | .27 | .71 | .71 | .55 | .32 | .72 | .72 | .55 | .32 |
| Frequency of applied ripple current | | 10 kHz | | | | 40 kHz | | | | 100 kHz | | | |
| Ambient still air temperature in °C | | ≤+55 | +85 | +105 | +125 | ≤+55 | +85 | +105 | +125 | ≤+55 | +85 | +105 | +125 |
| % of | 100% | .88 | .55 | - | - | 1.0 | .63 | - | - | 1.1 | .69 | - | - |
| +85°C | 90% | .88 | .67 | - | - | 1.0 | .77 | - | - | 1.1 | .85 | - | - |
| rated | 80% | .88 | .76 | .52 | - | 1.0 | .87 | .59 | - | 1.1 | .96 | .65 | - |
| peak | 70% | .88 | .85 | .64 | - | 1.0 | .97 | .73 | - | 1.1 | 1.07 | .80 | - |
| voltage | 66-2/3% | .88 | .88 | .68 | .40 | 1.0 | 1.0 | .77 | .45 | 1.1 | 1.1 | .85 | .50 |

1/ At +125°C, the rated voltage of the capacitors decreases to 66.67 percent of the +85°C rated voltage.
2/ The peak of the applied ac ripple voltage plus the applied dc voltage must not exceed the dc voltage rating of the capacitor either forward or reverse.
3/ The ripple current listed in table I represents a rating calculated by using a maximum internal temperature rise (ΔT) at +50°C at 40 kHz at +85°C ambient temperature, with a maximum peak rated voltage of 66.67 percent of the +85°C peak voltage rating.
4/ The maximum allowable internal temperature rise (ΔT) decreases linearly to a calculated 10°C rise at +125°C ambient.
5/ The internal temperature rise is directly proportional to the ESR of the capacitor, and ESR increases with decreasing frequency.



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Substitution of H designated parts for non H designated parts: The acquiring activity may substitute an H designated part in place of a non H designated part at their option.

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Referenced documents. In addition to MIL-PRF-39006, this document also references:

| | | |
|---|---|---|
| MIL-PRF-39006/25 | MIL-STD-202-112 | MIL-STD-202-214 |
| MIL-STD-202-101 | MIL-STD-202-204 | MIL-STD-202-301 |
| MIL-STD-202-105 | MIL-STD-202-208 | MIL-STD-202-302 |
| MIL-STD-202-106 | MIL-STD-202-210 | MIL-STD-202-305 |
| MIL-STD-202-107 | MIL-STD-202-211 | MIL-STD-810 |
| MIL-STD-202-108 | MIL-STD-202-213 | MIL-STD-1276 |

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2018-006)

Review activities:
Navy - AS, MC, OS
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